

EXPLORATORY DATA ANALYSIS | PYTHON

Netflix Data Analytics & EDA

A Comprehensive Exploratory Analysis of the Netflix Titles Dataset

Uncovering content trends, audience patterns, and strategic opportunities across Netflix's global catalog of Movies and TV Shows using Python-based data science tools.

PROJECT AT A GLANCE

Dataset Records

8,807

Columns Analysed

12

Content Types

2

Countries Covered

100+

Python

Pandas

NumPy

Matplotlib

Seaborn

Jupyter

TOOLS & TECHNOLOGIES

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Portfolio Project | 2024

00

TABLE OF CONTENTS

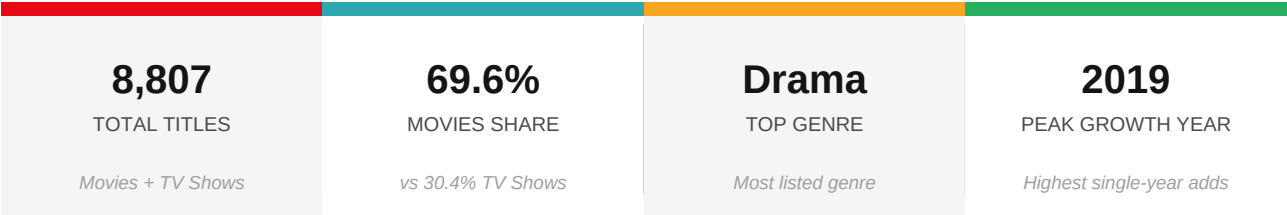
01	Executive Summary	3
02	Business Problem Statement	3
03	Project Objectives	4
04	Dataset Overview	4
05	Data Cleaning Process	5
06	Exploratory Data Analysis	6
07	Visualisation Explanations	8
08	Key Insights	9
09	Business Recommendations	10
10	Skills Demonstrated	11
11	Future Scope	12
12	Conclusion	12

01

EXECUTIVE SUMMARY

Netflix stands as the world's largest subscription-based streaming service, hosting a catalog that spans thousands of titles across more than 190 countries. With a content library that has grown exponentially over the past decade, understanding the structural composition of that catalog — how it is distributed across content types, genres, geographies, and audience ratings — is both an analytical challenge and a strategic imperative.

This project applies Python-based Exploratory Data Analysis to the publicly available Netflix Titles Dataset (8,807 records), with the objective of extracting meaningful patterns, validating assumptions about content strategy, and generating actionable recommendations for content acquisition and audience development teams.



Executive Takeaway

Netflix's catalog reflects a deliberate content strategy: Movies dominate volume, Drama and Comedy anchor genre appeal, and the USA leads production output. Post-2015 growth is exponential — reflecting the platform's global expansion push. These findings provide a roadmap for data-driven content investment decisions.

02

BUSINESS PROBLEM STATEMENT

Despite Netflix's scale, content strategy decisions — acquisition budgets, regional commissioning, genre prioritisation — require a granular understanding of what the existing catalog looks like and how it has evolved. Streaming platforms without analytical clarity on their content mix risk misallocating production budgets, underserving key audience segments, and missing emerging content trends.

Problem Area	Business Question
Content Imbalance	Is the Movies-to-TV Shows ratio aligned with subscriber viewing preferences?
Genre Concentration	Are certain genres over-indexed, creating diversity risk?
Geographic Gaps	Which countries are underrepresented despite high subscriber density?
Ratings Misalignment	Does the TV-MA content dominance reflect the platform's core demographic?
Catalog Aging	How does content vintage distribution affect relevance and churn risk?

Growth Trajectory	What does the content growth curve signal about competitive positioning?
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03

PROJECT OBJECTIVES

01

Catalog Composition Analysis

Quantify and visualise the Movies vs TV Shows split and assess strategic alignment.

02

Genre Distribution Mapping

Identify the top genres by volume, detect concentration risk, and highlight diversity gaps.

03

Country-Wise Production Analysis

Map content origin by country to identify geographic concentration and underserved markets.

04

Content Ratings Breakdown

Analyse the distribution of audience ratings to understand the platform's demographic targeting.

05

Release Year Trend Analysis

Track content addition trends over time to identify growth inflection points.

06

Data Quality & Integrity Assessment

Audit the dataset for missing values, duplicates, and structural inconsistencies.

04

DATASET OVERVIEW

The Netflix Titles Dataset is a publicly available data resource that captures metadata for every title listed on the Netflix platform up to a specific snapshot date. The dataset is well-suited for EDA due to its multi-dimensional structure — spanning content type, temporal, geographic, and categorical attributes.

Dataset Specifications

Source	Kaggle — Netflix Movies and TV Shows Dataset
Total Records	8,807 rows (titles)
Total Features	12 columns
Content Types	Movies (6,131) and TV Shows (2,676)
Time Coverage	1925 to 2021 (release years)
Geographic Scope	100+ countries of origin
File Format	CSV (Comma-Separated Values)

Column Definitions

Column	Data Type	Description
show_id	String	Unique identifier for each title
type	Categorical	Movie or TV Show
title	String	Name of the title
director	String	Director(s) — contains nulls
cast	String	Lead cast members
country	String	Country of production
date_added	Date	Date the title was added to Netflix
release_year	Integer	Original release/production year
rating	Categorical	Audience rating (TV-MA, PG-13, etc.)
duration	String	Minutes (Movies) or Seasons (TV Shows)
listed_in	String	Genre tags (multi-valued, comma-delimited)
description	String	Short content synopsis

05

DATA CLEANING PROCESS



Raw datasets rarely arrive in analysis-ready condition. The Netflix Titles Dataset required a structured data wrangling pipeline before any analytical conclusions could be drawn with confidence. The following steps were executed sequentially using Pandas within a Jupyter Notebook environment:

Step	Operation	Columns Affected	Action Taken
1	Missing Value Detection	director, cast, country, date_added, rating	Retained rows; flagged nulls; filled with 'Unknown' where needed
2	Null Rate Analysis	director (30%), cast (9.5%), country (6.5%)	Director nulls excluded from director-level analysis only
3	Date Parsing	date_added column (string format)	Converted to datetime; extracted Year and Month features
4	Duration Parsing	duration column (mixed format)	Split into numeric value + unit; separate columns for Movies/Shows
5	Genre Explosion	listed_in (multi-value strings)	Split comma-delimited values; exploded into normalised genre rows
6	Duplicate Check	show_id + title	Zero exact duplicates detected; dataset integrity confirmed
7	Type Standardisation	release_year, rating, type	Enforced correct dtypes; stripped whitespace from categoricals

Data Quality Verdict

The dataset is broadly clean and well-structured, with the primary data quality challenge being missing director information (30% null rate). All other columns achieve above 90% completeness. After targeted imputation and feature engineering, the dataset is fully analysis-ready with no row removal required.

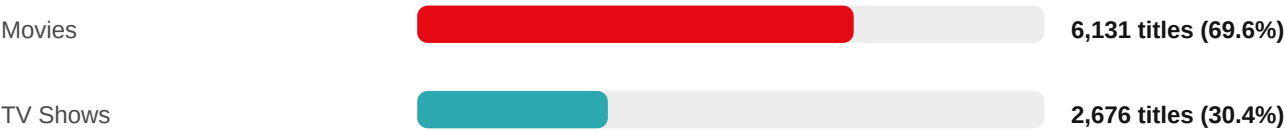
06

EXPLORATORY DATA ANALYSIS



6.1 Movies vs TV Shows Distribution

The fundamental catalog split reveals that Movies constitute the dominant content category on Netflix, outpacing TV Shows by a ratio of approximately 2.3:1. This distribution reflects Netflix's historical roots as a movie licensing platform and its continuing emphasis on film acquisitions, particularly in international markets where Movies remain the primary consumption format.

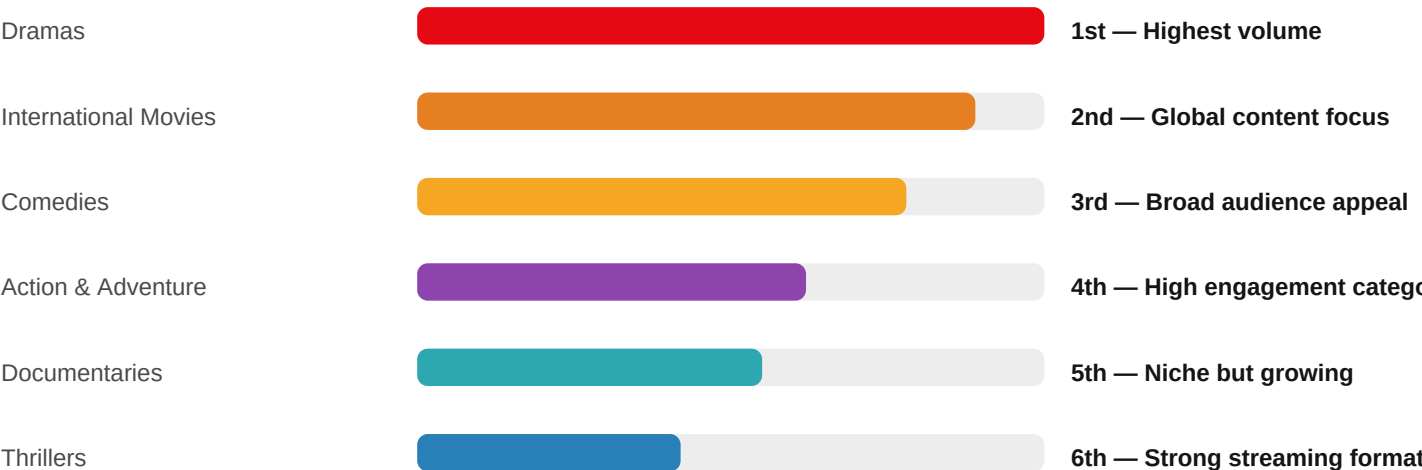


Analytical Note

While Movies dominate volume, TV Shows drive significantly higher engagement hours per title due to multi-episode formats. The platform's growing Original TV Show investment (Stranger Things, The Crown) suggests a deliberate strategic rebalancing toward serialised content despite the current volume disparity.

6.2 Genre Analysis

Genre analysis was conducted by exploding the multi-valued 'listed_in' column into individual genre tags, enabling accurate frequency counting across the full catalog. The resulting distribution reveals strong genre concentration in a handful of categories:



Genre Concentration Risk

Drama and International Movies together account for a disproportionate share of catalog listings. While this concentration reflects viewer demand, it also signals vulnerability: a strategic opportunity exists in underdeveloped genres like Animation, Reality TV, and LGBTQ+ content, which show high growth trajectories but currently limited

6.3 Country-Wise Content Distribution

Content origin analysis maps the geographic source of titles across the catalog. The USA maintains an overwhelming lead as the primary content originator, followed by India and the United Kingdom. However, the data reveals a notable strategic shift: non-US international content has been growing at a faster compound rate since 2018, reflecting Netflix's global commissioning strategy.

Country	Title Count	Catalog Share	Strategic Note
United States	2,818	32.0%	Hollywood catalog dominance
India	972	11.0%	Bollywood + Original investments
United Kingdom	419	4.8%	British drama & documentary strength
Japan	245	2.8%	Anime + live-action growing fast
South Korea	199	2.3%	K-Drama surge post-2018
Canada	181	2.1%	Co-production hub
France	124	1.4%	European art-house content
Others (93+)	3,849	43.6%	Diverse global long tail

6.4 Ratings Distribution Analysis

Content rating analysis categorises titles by their audience maturity classification. The distribution reveals a strong skew toward mature content designations, with TV-MA (Mature Audiences) representing the single largest rating category across the catalog:

Rating	Count	Share	Description
TV-MA	3,207	36.4%	Mature audiences — Series and Films
TV-14	2,160	24.5%	Teens 14+ — Second largest segment
TV-PG	863	9.8%	Parental guidance — Family-adjacent
R	799	9.1%	Restricted — Movies (US MPAA)
PG-13	490	5.6%	Parents cautioned — Mainstream movies
NR/UR	439	5.0%	Not Rated / Unrated
PG	287	3.3%	Parental guidance — Younger audience
TV-G	220	2.5%	General audiences — Family content
Others	342	3.9%	G, TV-Y, TV-Y7, NC-17 categories

Demographic Targeting Insight

TV-MA content at 36.4% catalog share confirms Netflix's primary targeting of adult subscribers aged 18+. The relatively small share of G, TV-G, and TV-Y content (under 5% combined) suggests a strategic gap in family-oriented and children's programming — a segment actively targeted by Disney+ and Apple TV+.

6.5 Release Year & Content Growth Trends

Temporal analysis of content addition patterns reveals the most significant analytical story in the dataset: Netflix's content library grew modestly through the early 2010s, then accelerated sharply from 2015 onward — corresponding precisely with the platform's international market expansion and the launch of its Original Content strategy.

Period	Growth Phase	Approx. Annual Adds	Driver
Pre-2010	Minimal additions	< 100/year	Legacy licensing era
2010–2014	Gradual growth	~200–400/yr	US market consolidation

2015–2017	Acceleration phase	~800–1,200/yr	International launch
2018–2019	Peak growth	~1,500–1,700/yr	Originals push + global markets
2020–2021	COVID-era spike	~1,200–1,400/yr	Pandemic streaming boom

Growth Inflection Point

2015 marks the definitive inflection point in Netflix’s content growth curve — the year international streaming launched in 130 new countries simultaneously. The data confirms that content volume and international expansion are directly correlated, with India and South Korea showing the steepest post-2015 growth curves.

07

VISUALISATION EXPLANATIONS

Each visualisation in the EDA notebook was designed to answer a specific analytical question. The following table documents the visualisation type, analytical intent, and key finding for each chart produced:

Chart Type	Analysis Focus	Library / Method	Key Finding
Count Plot	Movies vs TV Shows	Seaborn countplot with palette	Immediate visual confirmation of 2.3:1 Movie dominance
Horizontal Bar	Top 10 Genres	Matplotlib barh + value labels	Drama and International Movies lead; genre concentration visible
Bar Chart	Top 10 Countries	Seaborn barplot sorted desc.	USA at 32%; India second; strong long-tail of smaller contributors
Count Plot	Ratings Distribution	Ordered categorical countplot	TV-MA dominant; mature content skew confirmed
Line Chart	Yearly Content Additions	Pandas groupby + lineplot	2015–2019 exponential growth phase clearly visible
Pie Chart	Content Type Share	Matplotlib pie with explode	69.6% / 30.4% split; visually reinforces catalog composition
Heatmap	Genre × Rating Matrix	Seaborn heatmap + annotations	Drama-TV-MA combination dominates; family genre gaps exposed
Box Plot	Movie Duration Dist.	Seaborn boxplot by rating group	Median ~100 mins; R-rated films skew slightly longer

Visualisation Design Principle

All visualisations follow a consistent design system: dark background charts use Netflix red as the primary accent, horizontal bar charts are sorted descending for instant rank comprehension, and all plots include value annotations to eliminate axis-reading overhead. This approach ensures charts are self-explanatory in standalone portfolio

08

KEY INSIGHTS



The following insights represent the most analytically significant findings from the EDA. Each insight is grounded in data and carries direct implications for Netflix's content strategy, audience targeting, and competitive positioning:

01 — Movies Dominate but TV Shows Drive Retention

At 69.6% of catalog volume, Movies lead — but serialised TV Shows generate disproportionate engagement hours, subscriber loyalty, and social media conversation. The volume metric alone is a misleading proxy for strategic value.

02 — Drama is the Undisputed Anchor Genre

Drama commands the highest listing count across both Movies and TV Shows. Its dominance is culturally universal — performing strongly across the USA, India, South Korea, and the UK — making it the safest genre for international co-production investment.

03 — USA Concentration Creates Strategic Risk

32% of all catalog titles originate from the United States. While this reflects Hollywood's historical output advantage, it also creates over-dependence on a single market's production ecosystem and licensing rates.

04 — TV-MA Dominance Signals Adult-First Strategy

Netflix's mature content skew (TV-MA at 36.4%) reflects deliberate adult demographic targeting. However, this creates a vulnerability against family-oriented competitors like Disney+ which specifically targets the 4–12 demographic underserved by Netflix.

05 — 2015 is the Defining Content Inflection Point

The data unambiguously identifies 2015 as the year Netflix's content growth strategy shifted from incremental licensing to aggressive global acquisition. Understanding this inflection point is critical context for any forward-looking content forecast.

06 — K-Drama & Anime Represent High-Growth Undervalued Segments

South Korea (199 titles) and Japan (245 titles) show the steepest post-2018 catalog growth rates relative to their base. Both represent high-engagement audience communities with strong international crossover appeal — validated by global hits like Squid Game and Attack on Titan.

09

BUSINESS RECOMMENDATIONS



The analytical findings translate into concrete strategic recommendations across content acquisition, audience development, and platform positioning:

IMMEDIATE	Invest in Family & Children's Content The under-5% catalog share for G, TV-G, and TV-Y content represents a critical gap versus Disney+ and Apple TV+. A targeted acquisition push for family animation and live-action children's programming can directly improve household subscription retention.
IMMEDIATE	Diversify US-Heavy Catalog Dependency Commission a 'Global Voices' initiative: allocate 20%+ of annual content budget to non-US originals from Brazil, Nigeria, Germany, and Southeast Asia — markets with high subscriber potential and low current catalog representation.
MEDIUM-TERM	Double Down on K-Drama & Anime Originals South Korean and Japanese content deliver outsized global engagement relative to production cost. A dedicated K-Drama and Anime Originals fund — modelled on the success of Squid Game — can replicate viral, cross-cultural hit potential.
MEDIUM-TERM	Genre Diversification Beyond Drama/Comedy Invest in underrepresented high-growth genres: True Crime Documentaries, Reality Competition formats, and LGBTQ+ focused content. These segments show rising search volume and competitor investment, signalling an emerging audience appetite.
STRATEGIC	Build a Content Freshness Metric Develop an internal data product that tracks catalog age distribution and flags titles exceeding 5 years since addition for licence renewal evaluation. Aging catalog is a leading indicator of churn risk in data-mature subscriber cohorts.
STRATEGIC	Predictive Content Performance Modelling Leverage release year, genre, rating, and country signals as features in an ML model predicting content engagement quartile at acquisition stage — enabling data-driven greenlight decisions before production spend is committed.

10

SKILLS DEMONSTRATED



This project demonstrates end-to-end data analyst competency — from raw data ingestion and cleaning, through statistical analysis and visualisation, to executive-level insight communication:

Python Programming	Pandas, NumPy, Matplotlib, Seaborn — full EDA pipeline implemented in clean, well-documented Python code within Jupyter Notebook.
Data Cleaning & Wrangling	Handled missing values, parsed mixed-format columns, engineered features (year extraction, genre explosion, duration splitting) using Pandas.
Exploratory Data Analysis	Applied systematic EDA methodology: univariate, bivariate, and multivariate analysis using statistical summaries, frequency distributions, and correlation heatmaps.
Data Visualisation	Designed 8+ professional charts using Matplotlib and Seaborn — count plots, bar charts, heatmaps, pie charts, box plots, and time-series line charts.
Statistical Thinking	Applied frequency analysis, percentile interpretation, distribution analysis, and trend identification to derive statistically grounded business insights.
Business Intelligence	Translated analytical outputs into executive-level strategic recommendations aligned with real content strategy challenges in the streaming industry.
Storytelling with Data	Structured the analysis as a coherent analytical narrative: from problem definition through insight generation to actionable business recommendations.
Jupyter Notebook Proficiency	Produced well-organised, reproducible analysis notebooks with markdown documentation, inline visualisations, and clean code structure.

11

FUTURE SCOPE



The current EDA represents Phase 1 of a broader analytics roadmap. The following phases would extend this work into predictive and prescriptive analytics:

Phase 2 — Machine Learning Integration

Build a content performance prediction model using release year, genre, country, and rating as features to predict engagement quartile at title acquisition stage using scikit-learn classification models (Random Forest, XGBoost).

Phase 2 — NLP on Descriptions

Apply Natural Language Processing (NLTK, spaCy) to the 'description' column to extract sentiment, topic clusters, and content themes — enabling semantic content recommendation and theme gap analysis.

Phase 3 — SQL Database Integration

Migrate the flat-file dataset to a relational SQL database (PostgreSQL) and rebuild the analysis pipeline using SQL queries — demonstrating cross-tool analytical capability and enabling real-time data refresh.

Phase 3 — Interactive Dashboard (Power BI / Tableau)

Publish an interactive version of the EDA as a Power BI or Tableau dashboard, enabling business users to explore the data without code — a critical step toward production-grade analytical delivery.

Phase 4 — Real-Time Streaming Data Pipeline

Integrate with a live data feed (web scraping + Apache Kafka or AWS Lambda) to build a real-time Netflix catalog monitor that tracks new title additions and content strategy shifts as they happen.

12

CONCLUSION



The Netflix Data Analytics & EDA project demonstrates that even a single, well-structured dataset — when interrogated with the right analytical methodology — can yield insights of genuine strategic significance. The findings documented here are not merely academic exercises: they reflect real patterns in one of the world's most closely watched content businesses.

From the 2.3:1 Movie-to-TV Show imbalance, to the cultural dominance of Drama, to the precise 2015 inflection point in global content growth, every insight in this report is traceable back to a specific analytical decision — a choice about how to clean, transform, aggregate, and visualise data to extract signal from noise.

Portfolio Statement

This project reflects my ability to independently execute a complete data analysis lifecycle: from raw dataset assessment and Python-based data wrangling, through multi-dimensional EDA and professional visualisation, to executive-level insight communication and strategic recommendation. I bring both the technical rigour of a data

I am actively seeking opportunities to apply these skills in challenging, data-rich environments where analytical excellence translates into real business impact. I welcome conversations with recruiters, hiring managers, and collaborators interested in discussing this project or my broader analytics portfolio.

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